

Feature-relative residual gate — development comparison

Same 26 frozen failure-diagnostic objects, current checkpoint, six target views, and inference protocol.

One mechanism, two schedules: C3 + gate and Generic Linear + gate. Thresholds were calibrated from baseline GFL residuals before this run.

Method	n	CIEDE2000 ↓	FG-LPIPS ↓	FG-PSNR ↑	Edge-SSIM ↑	Lab flags
GFL	26	17.00	0.175	14.05	0.547	0/26
C3	26	18.78	0.175	13.82	0.547	0/26
Generic Linear	26	21.58	0.161	13.58	0.545	0/26
LLH	26	20.59	0.163	13.85	0.547	0/26
C3 + Feature Gate	26	20.14	0.166	13.68	0.538	0/26
Linear + Feature Gate	26	20.30	0.160	13.92	0.552	0/26

Primary endpoint is equal-weight mean per-view foreground CIEDE2000. Secondary metrics were recomputed from the saved RGB PNGs with the locked metric implementation.

All 26 cases are previously viewed or failure-enriched development diagnostics; these results are not independent confirmation. Purple threshold flags are shown with continuous color error and visual review.

The gate uses $g = \min(1, \tau_{\text{depth}} / q)$, $q = \text{RMS}(\text{scale-capped residual}) / \text{RMS}(\text{pre-adapter feature})$. It adds no trained parameters or network pass.

Following pages show full 3-row × 2-column RGB grids for each object. No image is cropped or recolored.

Paired six-view RGB comparison — objects 1-2 of 26

Reference

GFL

C3

Generic Linear

LLH

C3 + Gate

Linear + Gate

fig4_row1

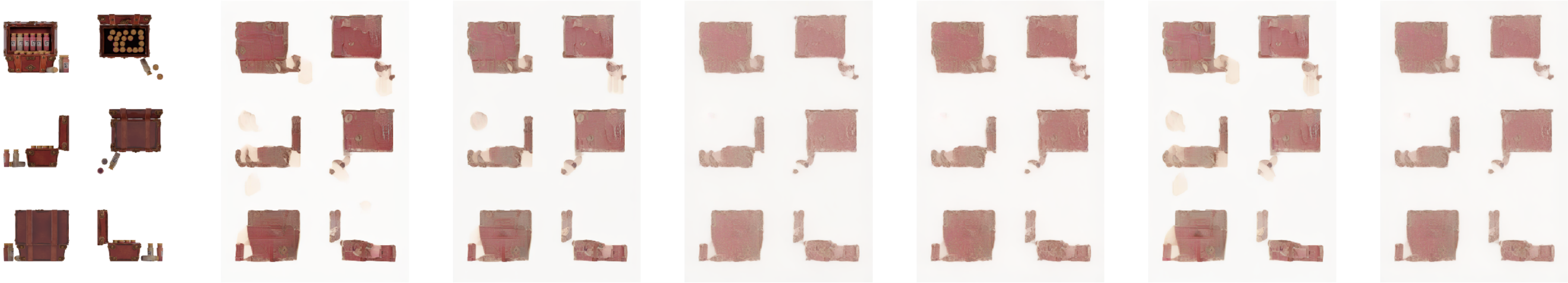


fig4_row2



Each panel is an unmodified 512×768 RGB grid containing the same six target views. Full assets and SHA-256 rows are in raw_rgb/ and logs/.

Paired six-view RGB comparison — objects 3-4 of 26

ReferenceGFLC3Generic LinearLLHC3 + GateLinear + Gate

fig4_row3

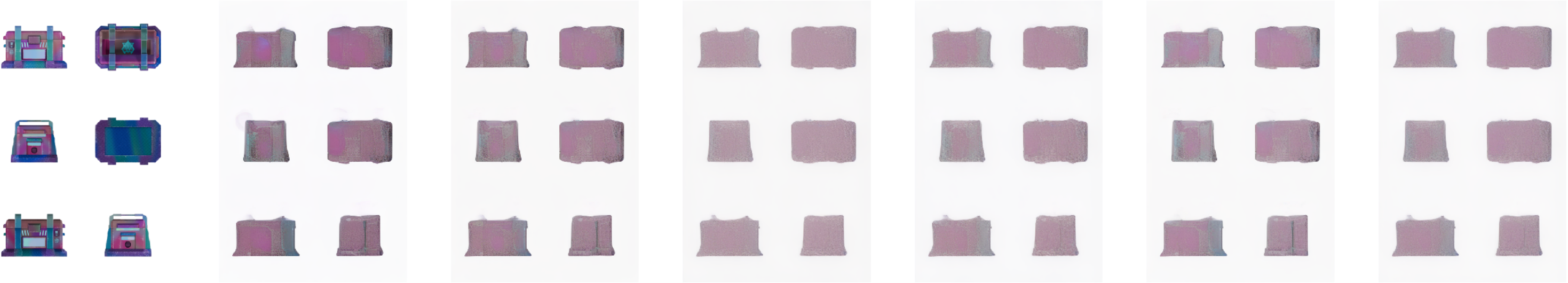
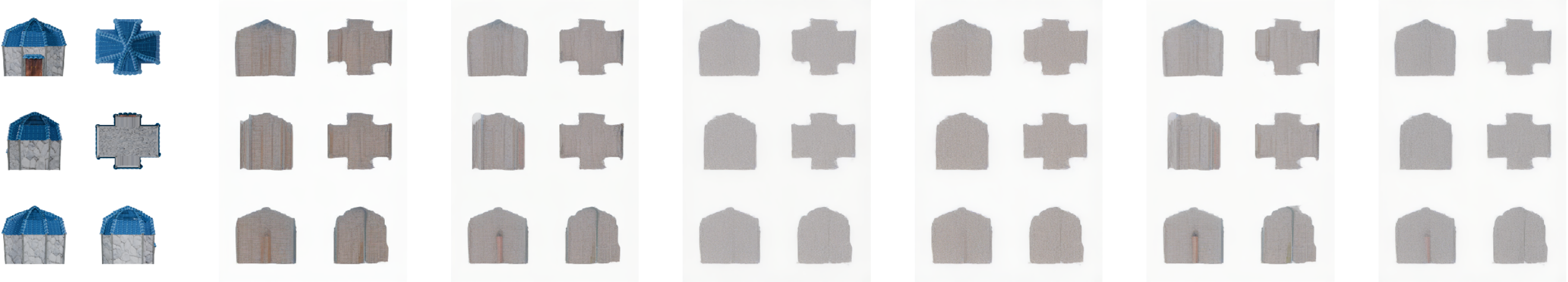


fig6_row1



Each panel is an unmodified 512×768 RGB grid containing the same six target views. Full assets and SHA-256 rows are in raw_rgb/ and logs/.

Paired six-view RGB comparison — objects 5-6 of 26

Reference GFL C3 Generic Linear LLH C3 + Gate Linear + Gate

fig6_row2

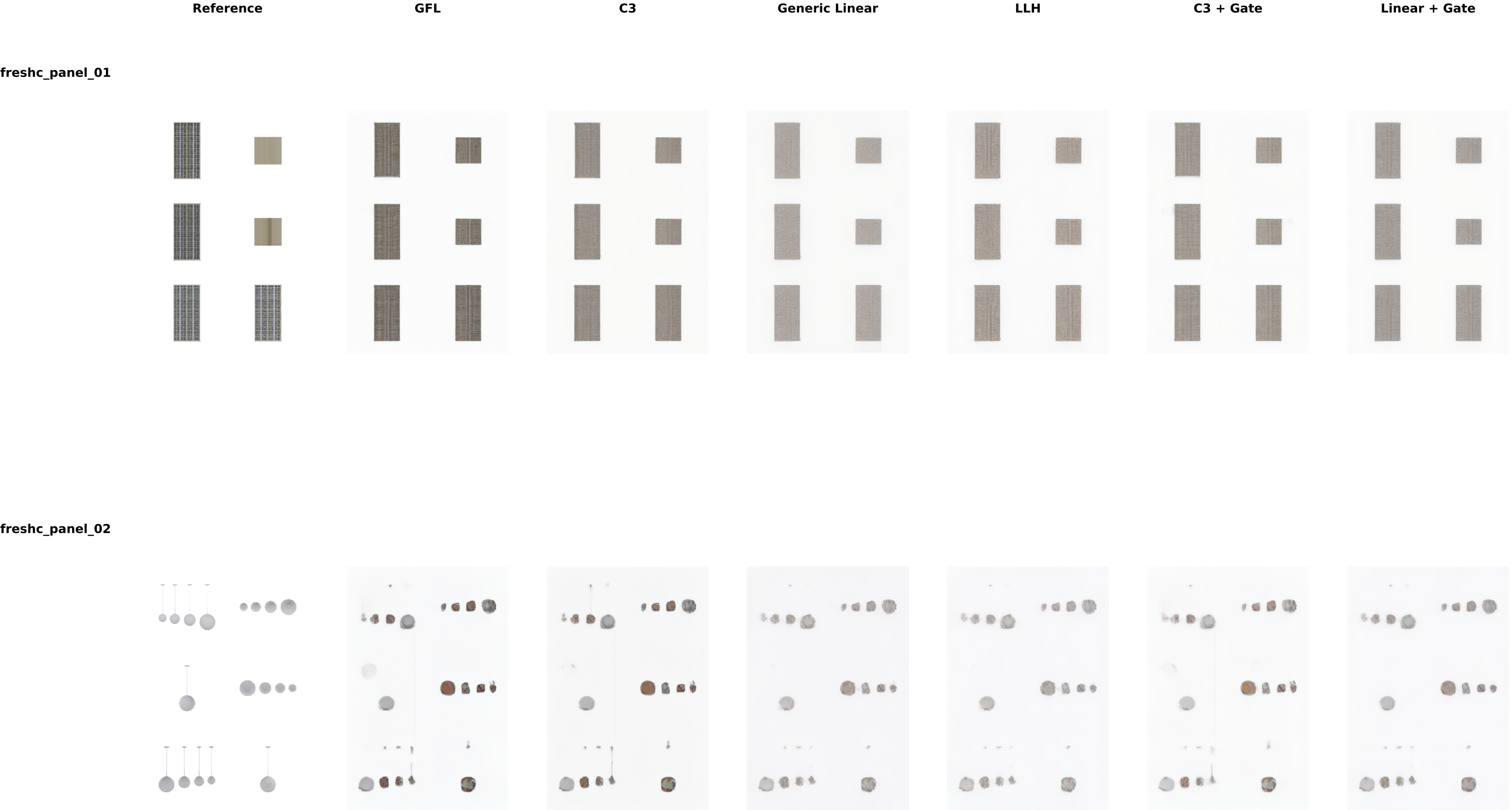


fig6_row3



Each panel is an unmodified 512×768 RGB grid containing the same six target views. Full assets and SHA-256 rows are in raw_rgb/ and logs/.

Paired six-view RGB comparison — objects 7-8 of 26



Paired six-view RGB comparison — objects 9-10 of 26

Reference

GFL

C3

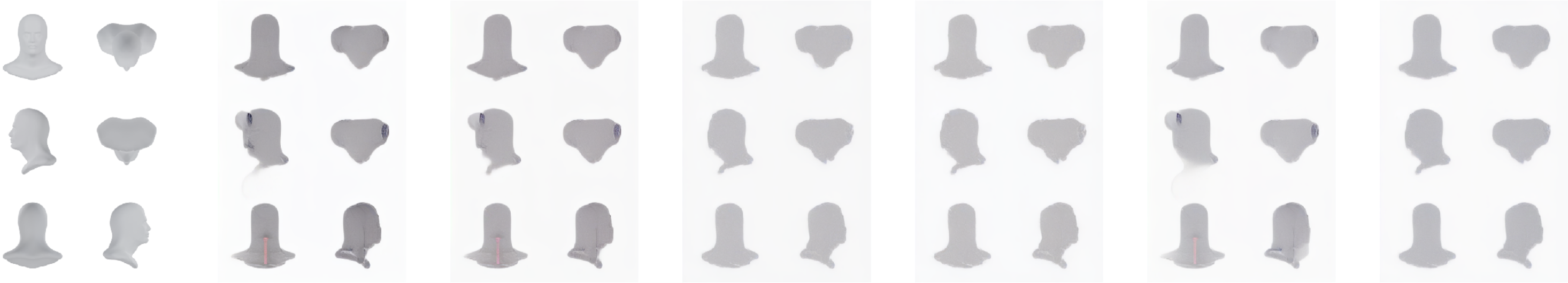
Generic Linear

LLH

C3 + Gate

Linear + Gate

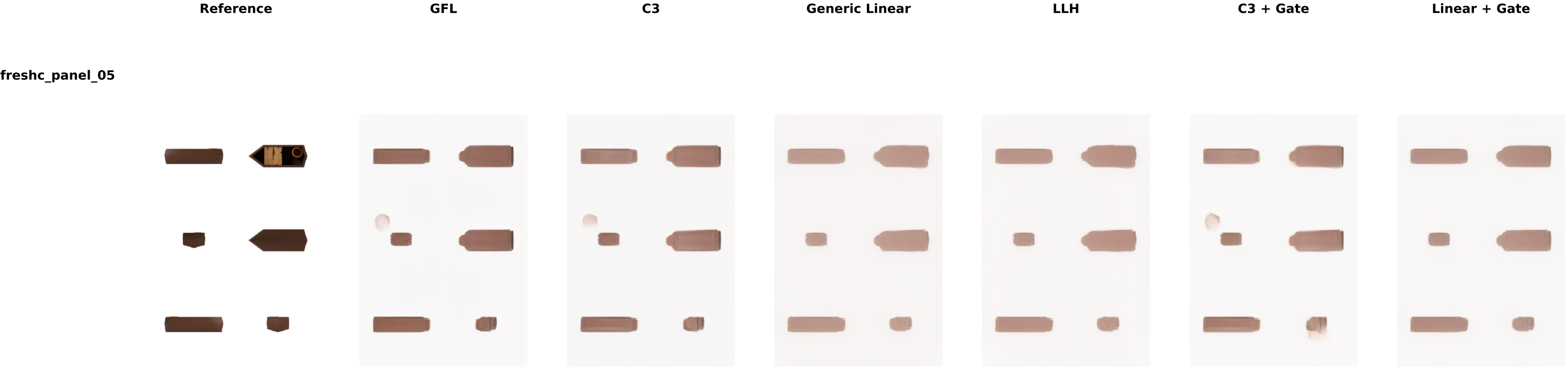
freshc_panel_03



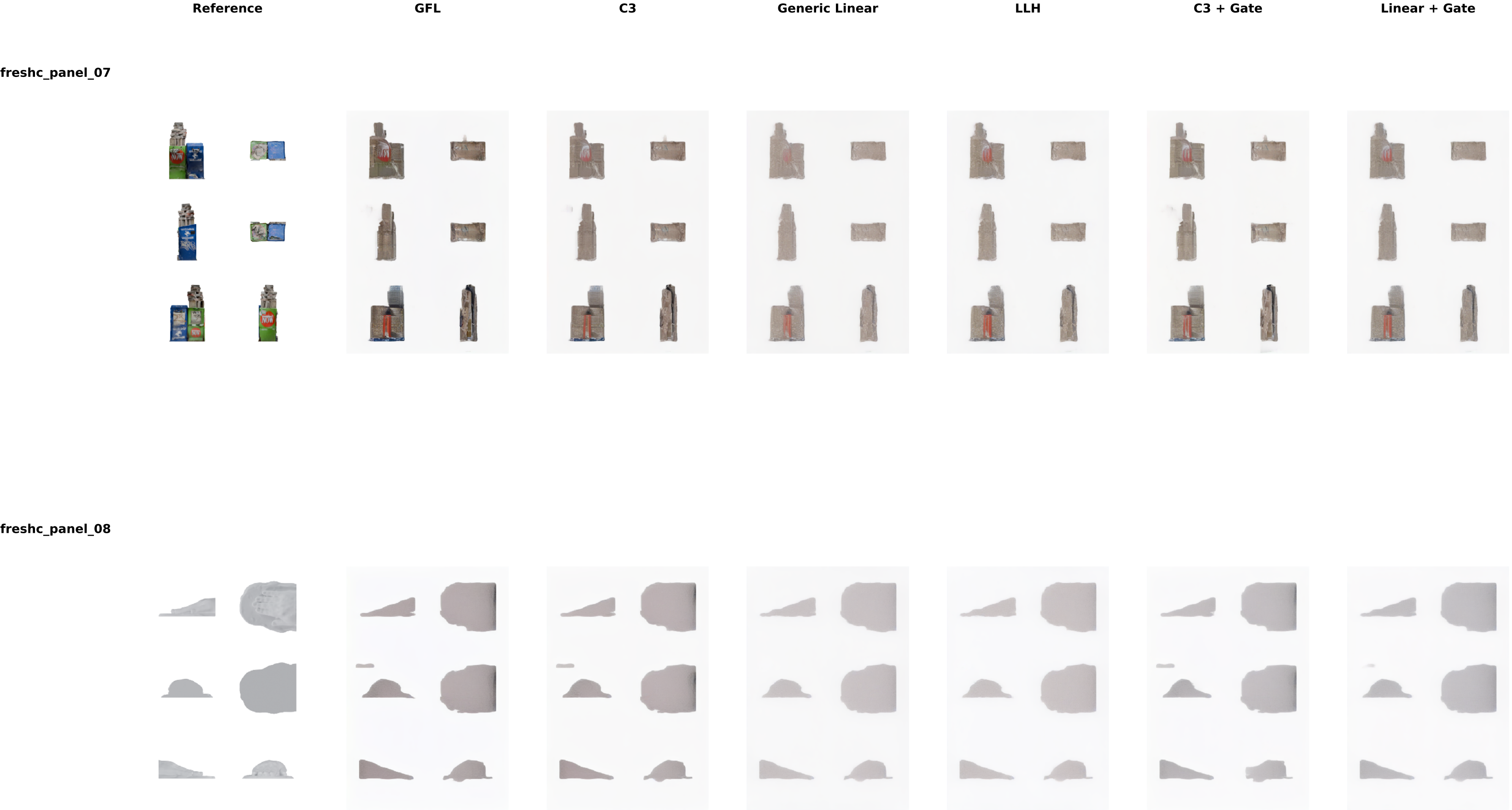
freshc_panel_04



Paired six-view RGB comparison — objects 11-12 of 26



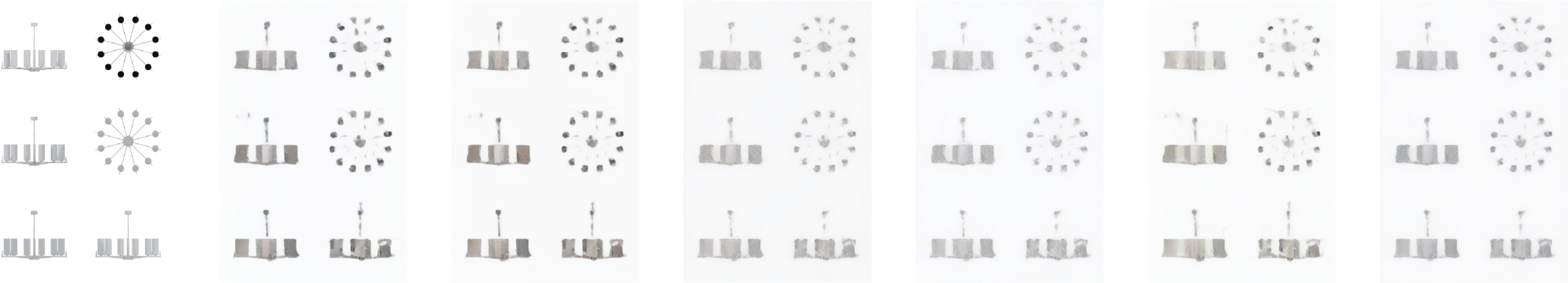
Paired six-view RGB comparison — objects 13-14 of 26



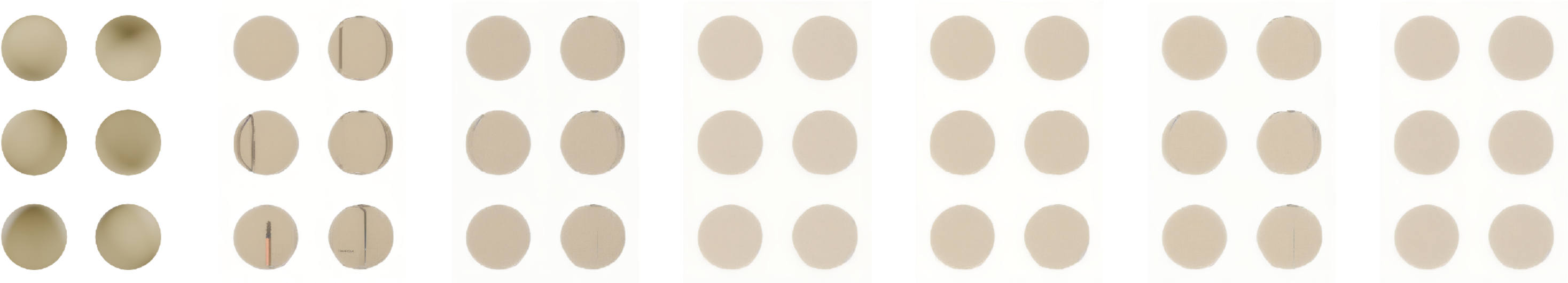
Paired six-view RGB comparison — objects 15-16 of 26

ReferenceGFLC3Generic LinearLLHC3 + GateLinear + Gate

freshc_panel_09



freshc_panel_10



Paired six-view RGB comparison — objects 17-18 of 26

Reference

GFL

C3

Generic Linear

LLH

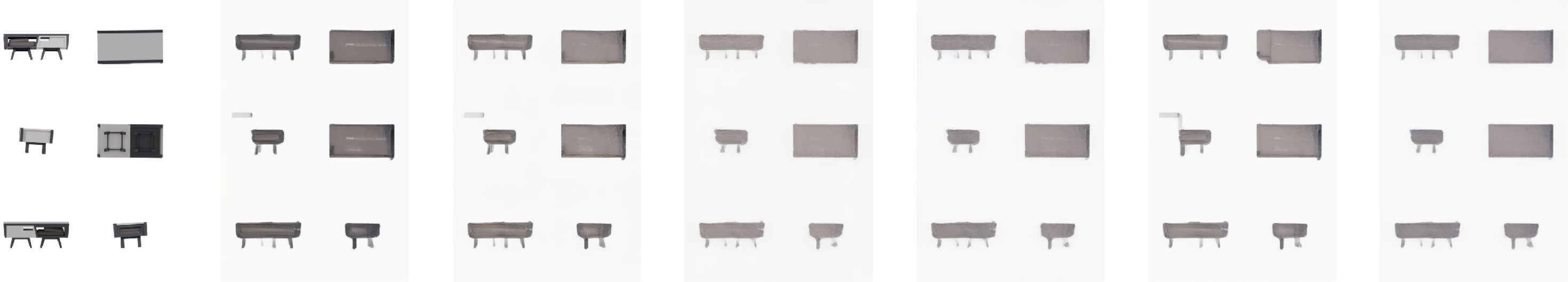
C3 + Gate

Linear + Gate

freshc_panel_11



freshc_panel_12



Paired six-view RGB comparison — objects 19-20 of 26

Reference

GFL

C3

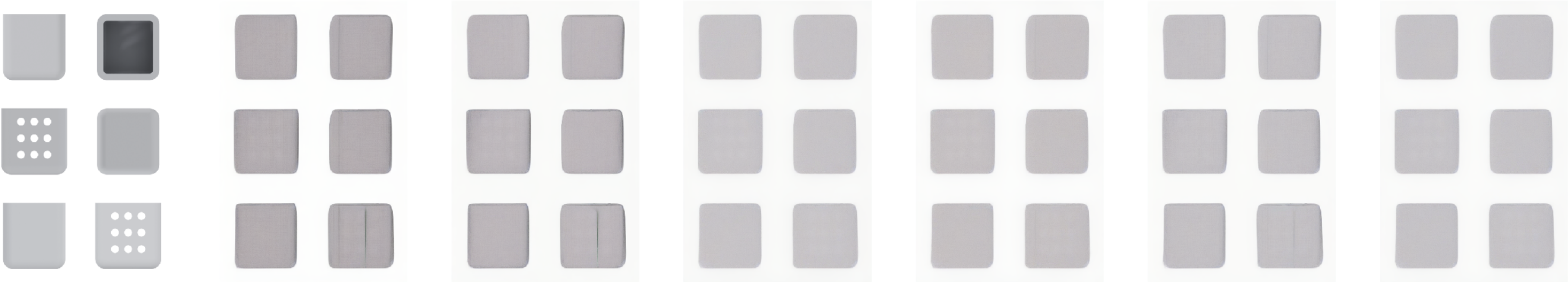
Generic Linear

LLH

C3 + Gate

Linear + Gate

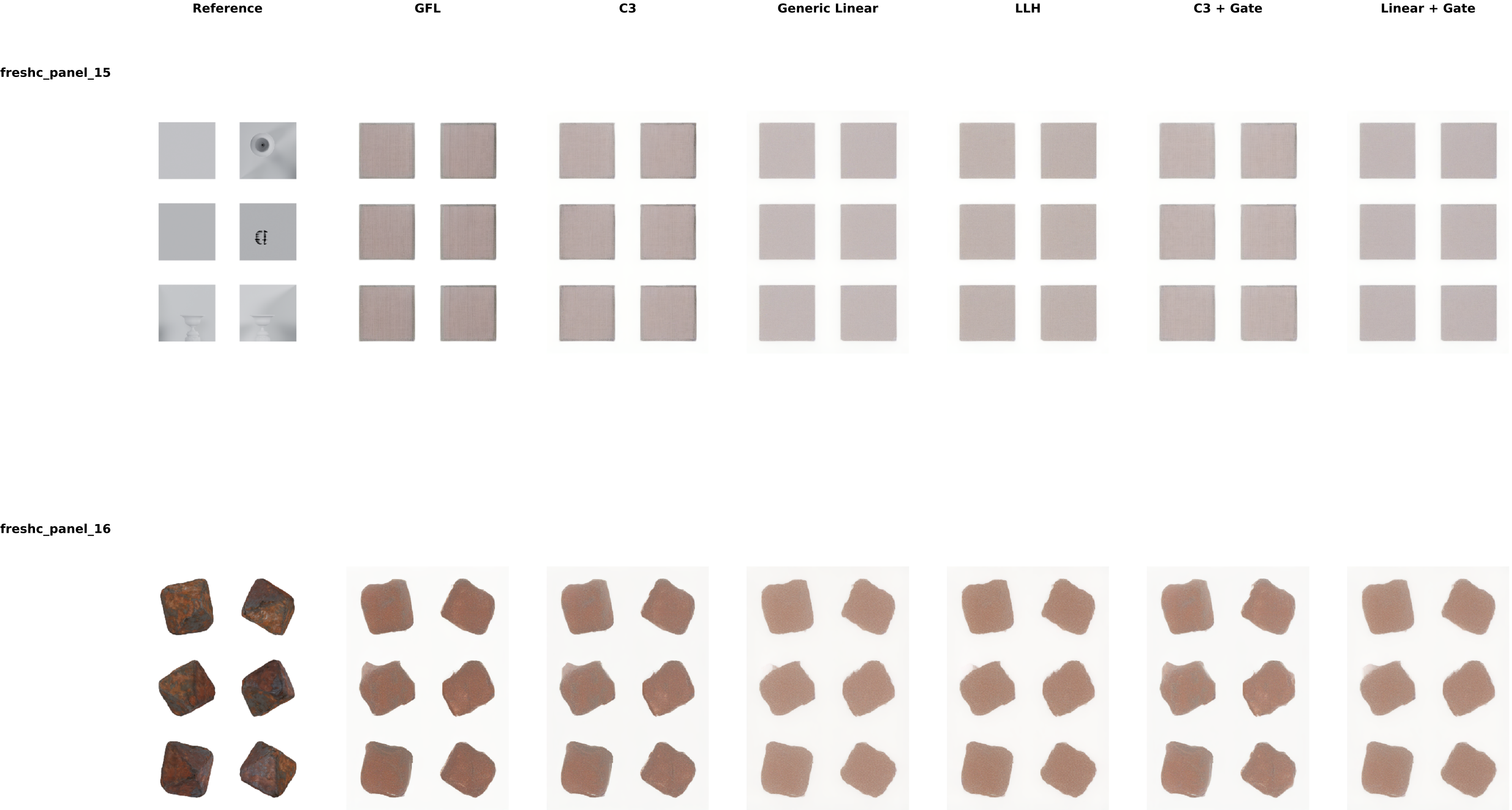
freshc_panel_13



freshc_panel_14



Paired six-view RGB comparison — objects 21-22 of 26



Paired six-view RGB comparison — objects 23-24 of 26

Reference

GFL

C3

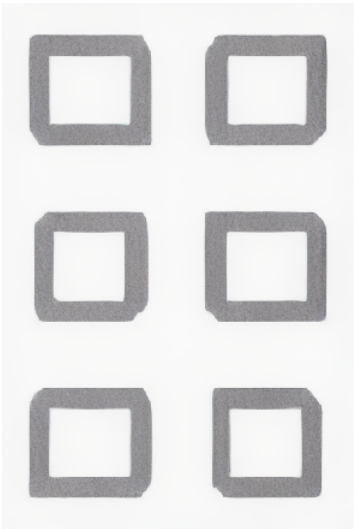
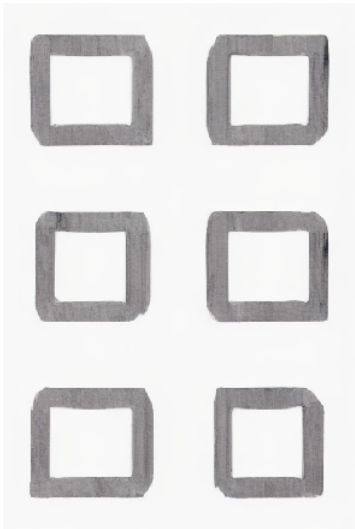
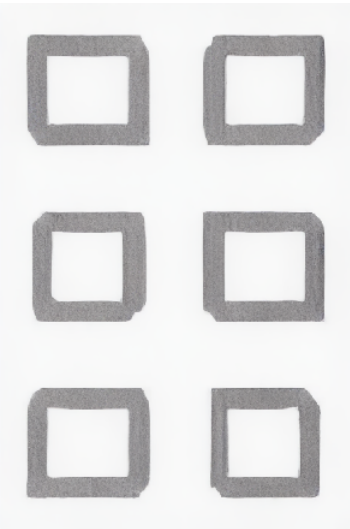
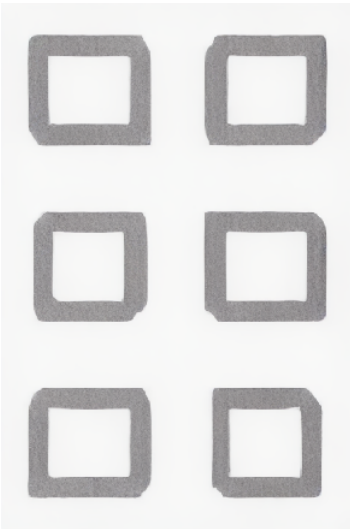
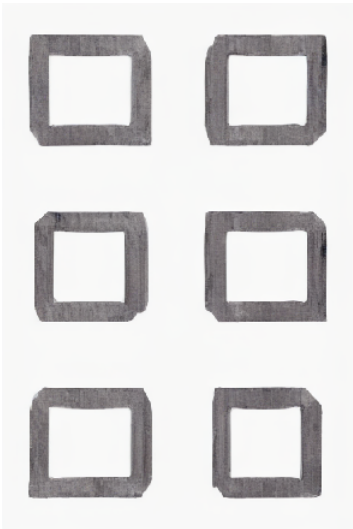
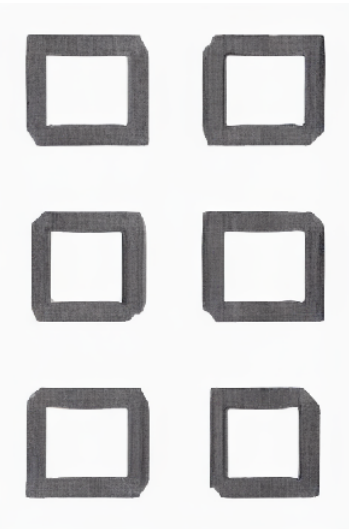
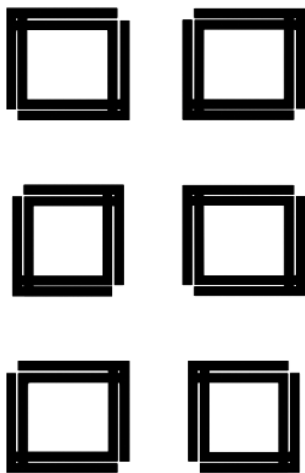
Generic Linear

LLH

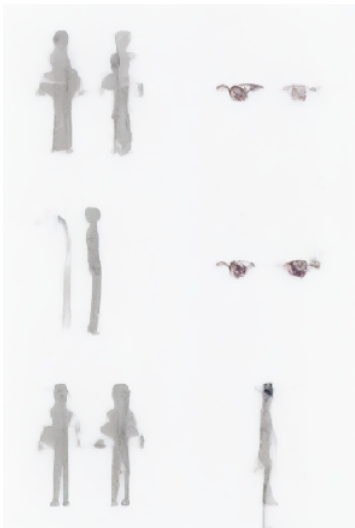
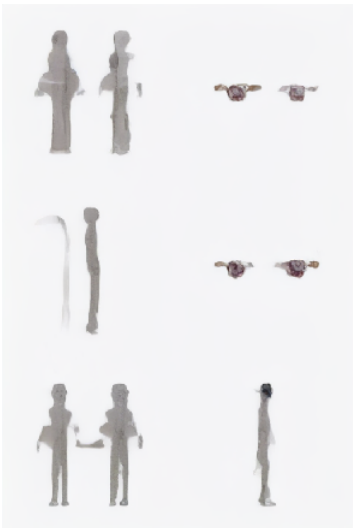
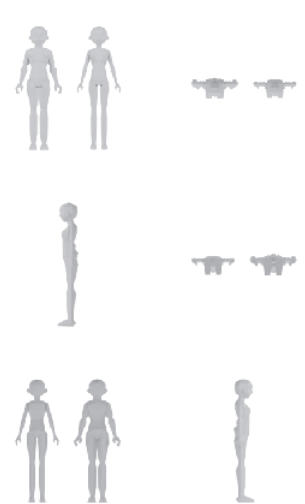
C3 + Gate

Linear + Gate

freshc_panel_17



freshc_panel_18



Paired six-view RGB comparison — objects 25-26 of 26

Reference

GFL

C3

Generic Linear

LLH

C3 + Gate

Linear + Gate

freshc_panel_19



freshc_panel_20

